

新加坡华乐团非常规乐器

SCO's Non-Standard Instruments

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1. 非常规华族打击乐器 Non-Standard Chinese Percussion Instruments

若需要用到以下乐器，请预先跟新加坡华乐团沟通。

Please consult SCO in advance before scoring for any of these instruments.

这些乐器音量不大，若使用，配器不宜太厚。

These are delicate instruments with a soft volume which can only be heard if the orchestration is thin.



编钟 Bianzhong



编磬 Bianqing



方响 Fangxiang

1.1. 编钟 Bianzhong

上层的音在中层也有，中层的音质更好，因此上层应避免使用。

The pitches of the top row are duplicates of the middle row. It is preferable to use the middle row (which sound better) and avoid the top row.

每个钟能发出一个正音和高小三度的侧音。多数钟上的正音比侧音发音更清楚。

Each bell produces a primary pitch and a secondary pitch a minor third higher. On most bells, the primary pitch can be sounded more clearly than the secondary pitch.

有些钟上敲侧音时，正音也会局部发音，造成难以避免的和声。

On some bells, striking the secondary pitch causes the primary pitch to sound partially too, resulting in an unwanted harmony that is hard to avoid.

最低音的 C3、D3、E3、F3 只能发出正音，因此从 A3 开始才有完整的半音阶。

The lowest C3, D3, E3 and F3 can only produce the primary pitch. Thus, the chromatic scale begins on A3.



若只使用少数的钟，演奏者会将其挂在更小的架子上，以便移动，也节省空间。

If a limited number of bells are used, these are usually mounted on a smaller rack to save space and for easier movement.

由于钟之间有些距离（尤其低音钟），可能需要两人以上演奏。具体看所使用的音高、音的顺序及速度等等。

The bells, especially the lower ones, are physically some distance apart. Depending on the pitches used and the speed and order of pitches played, more than one player might be needed.

1.2. 编磬 Bianqing (Stone chimes)

由于编磬已旧，有些磬片已经发音偏低或音色变质。特别是 E5、Eb5、Bb4、F4，已偏低将近半音。

As the Bianqing pieces are old, some of the pitches are out-of-tune (too flat) or have a suboptimal timbre.

Particularly, the E5, Eb5, Bb4 and F4 are flat by almost a semitone.

欲用编磬者，建议预先跟乐团安排测试乐器是否适用。

A composer wishing to use the Bianqing is recommended to first arrange a testing session with SCO to evaluate whether the instrument is suitable for their composition.

1.3. 方响 Fangxiang

较低音的方响基础音很弱，因此虽然在低音谱表音区，其音色仍然明亮。

The low Fangxiang pieces have weak fundamental frequencies, thus they produce a bright sound despite being in the bass clef register.

2. 东南亚打击乐器 Southeast Asian Percussion Instruments

这些均按半音阶定音。These are all tuned to the chromatic scale.



Gong ageng: 音高为 C1 的巨锣。A huge hanging gong pitched C1.

Kempul: 一套 13 个挂锣。最低的 (C2) 可当作 Gong Suwukan.

A set of 13 hanging gongs. The lowest (C2) can be used as the Gong Suwukan.

Bonang: 一套 13 个摆在平面上的罐锣。架子一次最多可排放 10 个锣。

A set of 13 kettle gongs placed on a flat rack. The rack can hold up to 10 gongs.

Angklung: 一套摇竹，需要预先安排所用的音需要几位演奏者。

A set of bamboo shakers. The number of players required for the pitches used should be arranged in advance.

3. 其它特殊打击乐器 Other Special Percussion Instruments

钟音板 (片钟): 新加坡华乐团有两片，音高为 G#2 和 B2.

Bell plates: SCO has two pieces, pitched G#2 and B2.

水晶磬: 新加坡华乐团的水晶磬音高为 Bb4，可倒水使其音高降低几个半音。

Crystal singing bowl: SCO's is pitched Bb4. Water can be poured into it to tune the pitch downwards by a few semitones. It can be struck or rubbed.

4. 传统笙 Traditional Sheng

传统笙有很多种制式。SCO 笙声部当中三位演奏家使用 D 调 26 簧笙（音列请见下图），一位高音笙及两位中音笙演奏家可兼奏。其余两位演奏家使用 D 调 21 簧笙，缺少括弧标记的音。不同的 21 簧笙的音列或有少许差别。如欲以传统笙在作品中扮演重要角色，作曲者可向 SCO 查询更多详情及指法表。

There are many models of traditional Sheng. Three of SCO's Sheng musicians use the 26-reed Sheng in D (pitches shown below). It can be doubled by one Gaoyin-Sheng musician and two Zhongyin-Sheng musicians. The remaining two Sheng musicians use the 21-reed Sheng, which lack the bracketed pitches. Some 21-reed Sheng have slight differences in the available pitches. If prominent parts are intended for the traditional Sheng, the composer may consult SCO for more details on the models available and their fingering charts.

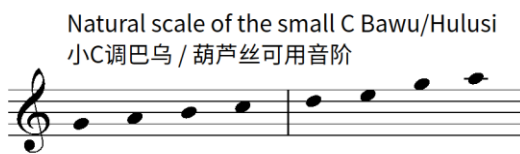


5. 其它特殊吹管乐器 Other Special Wind Instruments



埙可采用“低控音”技法，比最低音再吹出下约小三度音程

Xun can use the "dikongyin" technique to play around another minor third below the lowest pitch stated.



巴乌、葫芦丝市面上有 C、D、E $\flat$ 、E、F、G、A、B $\flat$ 调，请预先向 SCO 查询能否找到所用的调式。巴乌、葫芦丝虽也可演奏半音，但音色和音准则欠佳。

Bawu, Hulusi exist in C, D, E $\flat$, E, F, G, A, B $\flat$. Please check with SCO in advance whether the required key can be acquired. Although chromatic pitches outside the natural scale can be played, their timbre and intonation are suboptimal.

6. 其它特殊弦乐器 Other Special String Instruments

Please discuss any intention to use non-standard bowed string instruments (e.g., Banhu, Jinghu) or plucked string instruments (e.g., Qinqin or Yueqin) with SCO to see if the player and instrument can be organised.

若有意使用非常用弓弦乐器（如板胡、京胡）或非常用弹拨乐器（如秦琴、月琴），请预先与新加坡华乐团联系，以便探讨是否能安排演奏者及乐器。



照片 Photos



部分编钟放在更小的架子上
Selection of Bianzhong played on a smaller rack



两片方响
Two pieces of Fangxiang



各式木块（西式木鱼）、梆子、竹板
Wooden blocks, Bangzi, Zhuban (bamboo clappers)



大木鱼
Large Muyu



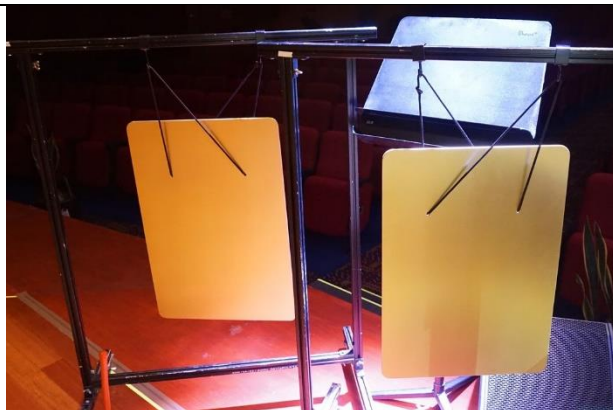
Bonang (rack can hold 10 of 13 kettle gongs)
罐锣（共 13 个，架子一次可放 10 个）



Kempul (6 of 13 shown)
挂锣（共 13 个，这里挂了 6 个）



Ceng-ceng (Balinese cymbals) 巴厘小镲



Bell plates 钟音板（片钟）